

Algorithms



Lecture 6

Karatsuba for Multiplication (Divide and Conquer)

Prantik Paul [PNP]

Lecturer

Department of Computer Science and Engineering

BRAC University

Naive Algorithm

$$\begin{array}{r} 12 \\ \times 34 \\ \hline 48 \\ 360 \end{array}$$

$$2 * 4 = 8$$

$$1 * 4 = 4$$

$$2 * 3 = 6$$

$$1 * 3 = 3$$

4 multiplications required

Naive Algorithm

X = 1234

Y = 5678

1234 x 5678

Divide and Conquer Approach

$X = X_1 * 10^{n/2} + X_r$ [X_1 and X_r contain leftmost and rightmost $n/2$ bits of X]

$Y = Y_1 * 10^{n/2} + Y_r$ [Y_1 and Y_r contain leftmost and rightmost $n/2$ bits of Y]

$$\begin{aligned} XY &= (X_1 * 10^{n/2} + X_r) (Y_1 * 10^{n/2} + Y_r) \\ &= 10^n X_1 Y_1 + 10^{n/2} (X_1 Y_r + X_r Y_1) + X_r Y_r \end{aligned}$$

$$X = 1234$$

$$Y = 5678$$

$$X_1 = 12, X_r = 34$$

$$Y_1 = 56, Y_r = 78$$

Divide and Conquer Approach

$$X = X_1 * 10^{n/2} + X_r \quad [X_1 \text{ and } X_r \text{ contain leftmost and rightmost } n/2 \text{ bits of } X]$$

$$Y = Y_1 * 10^{n/2} + Y_r \quad [Y_1 \text{ and } Y_r \text{ contain leftmost and rightmost } n/2 \text{ bits of } Y]$$

$$\begin{aligned} XY &= (X_1 * 10^{n/2} + X_r) (Y_1 * 10^{n/2} + Y_r) \\ &= 10^n \boxed{X_1 Y_1} + 10^{n/2} (\boxed{X_1 Y_r} + \boxed{X_r Y_1}) + \boxed{X_r Y_r} \end{aligned}$$

We have divided the entire problem to 4 subproblems

Still same 4 multiplications required, not an optimization

Divide and Conquer Approach

$$X = X_1 * 10^{n/2} + X_r \quad [X_1 \text{ and } X_r \text{ contain leftmost and rightmost } n/2 \text{ bits of } X]$$

$$Y = Y_1 * 10^{n/2} + Y_r \quad [Y_1 \text{ and } Y_r \text{ contain leftmost and rightmost } n/2 \text{ bits of } Y]$$

$$\begin{aligned} XY &= (X_1 * 10^{n/2} + X_r) (Y_1 * 10^{n/2} + Y_r) \\ &= 10^n X_1 Y_1 + 10^{n/2} (X_1 Y_r + X_r Y_1) + X_r Y_r \end{aligned}$$

Change this term to something better

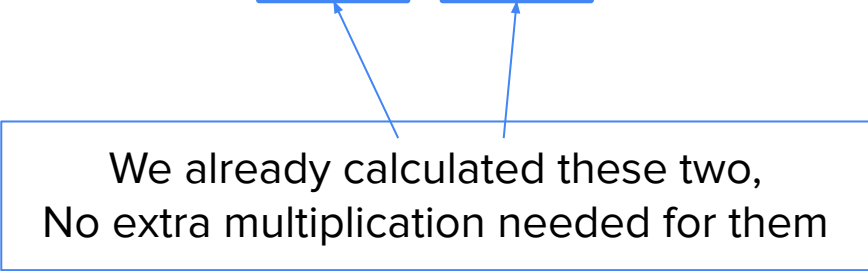
Divide and Conquer Approach

Tricky expression for middle two terms:

$$X_1 Y_r + X_r Y_1 =$$

Divide and Conquer Approach

Tricky expression for middle two terms:

$$X_1 Y_r + X_r Y_1 = (X_1 + X_r) (Y_1 + Y_r) - \boxed{X_1 Y_1} - \boxed{X_r Y_r}$$
A diagram consisting of a rectangular box at the bottom containing text. Two blue arrows originate from the top of this box. One arrow points diagonally upwards and to the left, ending at the bottom of the first boxed term, $X_1 Y_1$. The other arrow points diagonally upwards and to the right, ending at the bottom of the second boxed term, $X_r Y_r$.

We already calculated these two,
No extra multiplication needed for them

Divide and Conquer Approach

Tricky expression for middle two terms:

$$X_1 \boxed{\times} Y_r + X_r \boxed{\times} Y_1 = (X_1 + X_r) \boxed{\times} (Y_1 + Y_r) - X_1 Y_1 - X_r Y_r$$

Instead of these two
multiplications

Only one new multiplication
needed

Divide and Conquer Approach

So the final value of XY becomes

$$XY = 10^n \boxed{X_1 Y_1} + 10^{n/2} \boxed{(X_1 + X_r)(Y_1 + Y_r)} - X_1 Y_1 - X_r Y_r + \boxed{X_r Y_r}$$

3 multiplications required instead of 4

$$\begin{aligned} &12 \times 56 \\ &= (12 \times 56) \times 10^4 \\ &\quad + ((12 + 34) \times (56 + 78) - 12 \times 56 - 34 \times 78) \times 10^2 \\ &\quad + 34 \times 78 \end{aligned}$$

Divide and Conquer Approach

Analyzing time complexity

$$XY = 10^n X_1 Y_1 + 10^{n/2} (X_1 Y_r + X_r Y_1) + X_r Y_r$$

$$T(n) = 4T(n/2) + \Theta(n)$$

=

Divide and Conquer Approach

Analyzing time complexity

$$XY = 10^n X_1 Y_1 + 10^{n/2} ((X_1 + X_r)(Y_1 + Y_r) - X_1 Y_1 - X_r Y_r) + X_r Y_r$$

$$T(n) = 3T(n/2) + \Theta(n)$$

$$= \Theta(n^{\log_2 3}) \text{ (master th'm)}$$

$$= \Theta(n^{1.59}) \text{ which is less than } \Theta(n^2)$$